AWS Research Paper

By

Timothy Bates

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Architecture and Security

Dr. Mahd Alzoubi

University of West Florida

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Abstract

This paper explores the design, implementation, and security constraints of cloud computing and its impact on modern IT infrastructure. There are several formats for cloud computing and implementing it into a business; software as a service, platform as a service, and infrastructure as a service are the primary models. This research paper addresses cloud computing security and how technologies and services support the principles of the C.I.A. triad. Amazon Web Services, a leading cloud service provider, will be the primary focus for how this is achieved in real-world examples.

Introduction

Cloud computing is a groundbreaking technological advancement and has become a pivotal part of IT infrastructures while revolutionizing the modern world. At its core, cloud computing entails the distribution of various computer-related resources over the internet. This offers virtualization options for a variety of services and hardware, such as: databases, networking, storage, servers, software, and analytics. This innovation allows companies access to flexible resources that can be customized to suit their business needs. In the past, IT infrastructures were an expensive undertaking for a company; from the cost of physical hardware to the overhead and cost of securing, maintaining, and updating these systems. Now, the ability to access this kind of infrastructure on an as-needed basis gives companies more flexibility and scalability, while lowering operating costs and leading to a more efficient infrastructure. The benefits of cloud computing extend beyond infrastructure needs into the realm of software development. Companies that develop software applications can utilize flexible, scalable, and efficient platforms to develop and run their applications without having to focus on hardware requirements and overhead. While companies that have software needs can utilize cloud hosted software applications directly as a service without needing to spend the capital to create them themselves. The flexibility of the cloud allows it to be utilized in a variety of industries for a wide variety of use-cases. Overall, there are several companies that offer cloud computing services, but for the purpose of this research paper, Amazon Web Services (AWS) and its diverse range of products will be the primary focus regarding examining cloud models, usage, and security concerns.

Research Questions

The wide scope of cloud computing leading to its implementation in an even wider array of industries has made cloud security a top priority. There are several pressing cloud data security issues that will be condensed to three areas of focus for further research: confidentiality and integrity of data, cloud data compliance and regulatory adherence, and external threats leading to availability concerns.

The first of these research questions is how to mitigate confidentiality and integrity risks of data stored on the cloud. Cloud environments make it easy to share data within them. This is a major key to the collaborative aspects of the cloud; however, concerns arise with the possibility of data leakage. 69% of organizations pose this as the greatest security concern for their cloud infrastructures. (Checkpoint,2024) Multiple tenants sharing the same cloud service, which is the reality for most organizations, makes the unauthorized access of data more of a possibly and thus a larger concern. System misconfigurations and data breaches also contribute to data loss, being a main security concern.

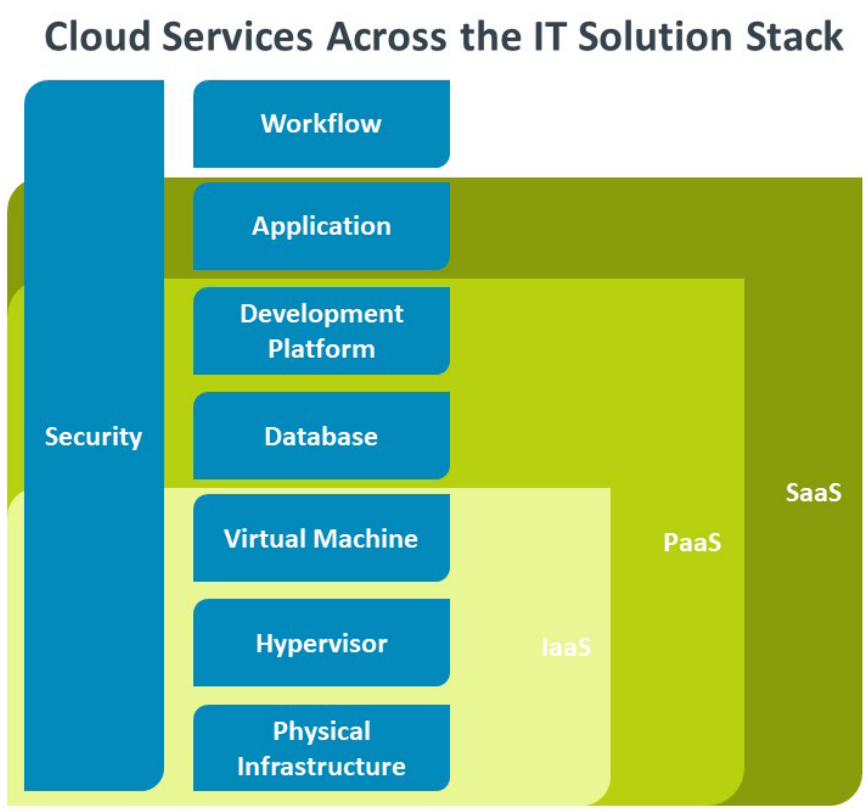
The second main security concern and research question is the how to achieve and maintain data compliance and regulatory adherence for cloud environments. The wide application of cloud services to different industries means that adhering to the specific regulatory requirements for data handling and storage for those industries becomes a critical priority for secure operations in those industries. Improper and insecure use of the cloud can lead to sensitive data being exposed exacerbating data loss issues as well. Ensuring compliance with major regulations can be complex and challenging due to the reality of data stored in a cloud environment, possibly being subject to multiple areas of geographic jurisdiction and legal requirements. These issues lead to nearly 42% of organizations requiring special cloud compliance solutions. (Checkpoint,2024).

The third research area for cloud security issues is the prevalence of cyberattacks, the resulting data availability concerns, and hardening those cloud infrastructures. Cloud services are essential for many companies' abilities to conduct business and maintain daily operations. Critically important information stored on the cloud and internal/customer facing software applications depend on this infrastructure being highly available. Due to this dependency, denial of service (DoS) attacks against this vital cloud infrastructure can have a major impact on the companies that depend on them. Other forms of cyberthreats can also contribute to data leakage as sensitive information can be exposed for threat agents inside and outside the organization.

Overall, these concerns regarding security aspects of cloud infrastructure pertain to a few fundamental issues. In a multi-tenant environment and simply due to the nature of data handling data loss and data exposure are a risk. Regulations and legal adherence to the jurisdiction of governing bodies due to the varied and globalized demand of cloud services. Also, the security and resilience of the cloud environment from external threats and attacks. These issues have a variety of solutions and mitigation strategies to help secure data company data within a cloud environment and keep the infrastructure available and efficient for business needs. The purpose of this research is to detail and explore the different components of cloud computing models and their architecture and how their advantages and disadvantages contribute to the outlined security concerns. The questions raised by these security concerns and their possible solutions will be explored as implementation may vary among these different cloud service models.

Cloud Models and Security Concerns

Figure 1



https://www.comptia.org/content/articles/what-is-paas

Software as a Service

There are multiple types of cloud models used by organizations to suit their business needs and requirements, the most streamlined of which is the Software as a Service (SaaS) model. SaaS applications are hosted by the cloud vendor and available to users and organizations on the internet. SaaS use cases are broad and comprise any enterprise or personal services for software applications such as email, sales management, file hosting platforms, and financial management. Some real-world examples of SaaS are Adobe Creative Cloud, Dropbox, DocuSign, and Slack. (TechTarget,2021) For this model businesses don't have to download or install software to their existing IT infrastructure and can simply access the required resources the vendor hosts. The vendors often offer a subscription to their users broken down by software requirements and user base.

Advantages

SaaS offers many benefits to the organizations that rely on them. One of the core advantages is that organizations can often gain immediate access to the services they need, and the software is streamlined to set up and operate. The responsibility of software management and maintenance is placed on the vendor making SaaS the most economical cloud model for an organization. The vendor handling these aspects ensures that the often-essential software for an organization is always supported, maintained, and up to date security wise. Scalability also becomes another benefit for the client organization as the hardware requirements to be able to run this software and a desired performance level does not cost the client. Additionally, not having to spend development time to create the software to meet that organization needs is another cost benefit that is simply bundled into the subscription of the service.

Disadvantages and Security risks

Though SaaS is an economical and beneficial choice for an organization there are several security related disadvantages to be aware of. Since the client organization doesn't have direct access to their vendor's cloud infrastructure any outages and service interruptions, for example, natural disasters or denial of service attacks (DoS) are often critical to the client organization's business operations. SaaS ecosystems for an organization can often comprise multiple applications from multiple vendors as each piece of software serves a different role. However, this can often create a fragmented platform that potentially leads to gaps in defense and threat monitoring capabilities. Some of these gaps can also be from human error, as misconfigurations can become more likely in a complex custom environment. These gaps can become easily exploitable weaknesses for cybercriminals and lead to data breaches. (TechTarget,2021)

Platform as a Service

Platform as a service (PaaS) offers developers a stage for software development and deployment over the internet. This gives developers easy access to powerful and up-to-date tools in a more economical fashion for their organization. The PaaS cloud vendor manages the servers, storage, and networking, while the business developers manage their software applications. PaaS versatility is not as broad as SaaS industry wise but still supports valuable use cases. It can support API development and the utilization of a wide scope of programming languages and application environments. It even has use in business analytics for data management and decision-making. Some real-world examples of PaaS are AWS Elastic Beanstalk, Google App Engine, AWS Lambda, and Microsoft Azure. (TechTarget,2021)

Advantages

Similarly to SaaS, and cloud computing in general, PaaS has a core benefit of providing an economical, scalable, and versatile service that users can access from the internet. However, unlike the ready to use hosted software of SaaS models, PaaS provides IT infrastructure and IT tools and services for development. PaaS platforms give organizations necessary compute and storage infrastructure for developers to collaborate on and deploy new software products. This is often supported with testing and compiling services that offer version control management to support best software development practices.

Disadvantages and Security risks

Some of the disadvantages of PaaS stem from its differences from SaaS in implementation. Often the tools and services of PaaS can be rigid and less scalable to meet high demands, this can cause resource stalls for the organizations that depend on them. Additionally lock-in can become an issue, which is businesses getting stuck in their vendor's platform and tools even when those resources might not meet the company's business requirements. From a security perspective, the responsibility of implementing and maintaining security standards and best practices is shared between the cloud vendor and the client business. The vendor is responsible for the infrastructure and platform. While the business is responsible for securing the applications they develop on the platform. This means that platform and application security deficiencies can lead to vulnerabilities that attacks can exploit to expose sensitive data. Critically, application vulnerabilities, which the cloud vendor has no part in securing in PaaS, can be a vector for affecting the security of the entire PaaS ecosystem. (TechTarget,2021)

Infrastructure as a Service

Infrastructure as a service is a model utilized by companies that either could not afford or do not prioritize maintaining on premise IT infrastructure. The cloud vendor hosts hardware components such as servers, storage, and networking. They are also responsible for the virtualization layer; however, the tenant handles all other aspects of the environment. This cloud model provides the most flexibility for the user regarding customizing their environment and the software application and tools they have the ability to host. However, this model also places the most responsibility on the user as the cloud vendor does not play a part in securing the development tools and applications.

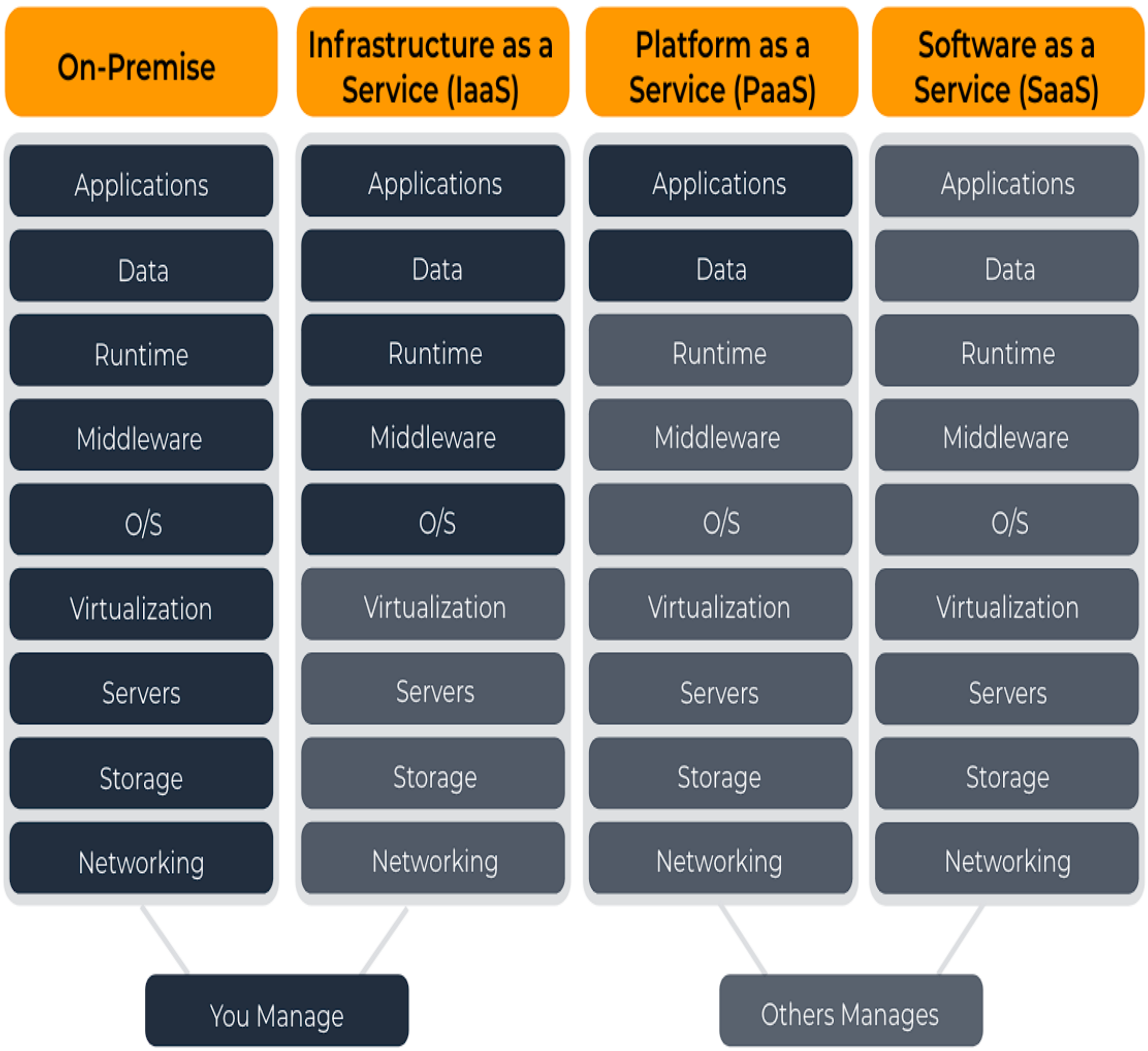
Advantages

The benefits of IaaS are primarily to offer enhanced scalability and resource flexibility for an organization in a customizable economical fashion. This can be a company that is expanding rapidly and cannot bring in physical hardware to support the expansion. It can also be a benefit for companies looking to save money and overhead by virtualizing some of their infrastructure. The pay-as-you-go model allows users to only pay for what they use and allows for dynamic scaling up and down to meet business needs thereby increasing efficiency. The cloud vendor handling security, outage recovery, updates, and routine maintenance means the virtualized hardware the user gets access to will be highly available, highly secure, and up to date with no proportional cost to the user organization. (Spot,2024)

Disadvantages and Security risks

Similar to the other two cloud models covered thus far IaaS is not without its disadvantages. Security and reliability being in the hands of the cloud vendor are a net positive for the user organization in terms of overhead and cost when functioning as intended. However, the lack of control the user has over these aspects can actually become a disadvantage. Not all vendors are created equally and not all issues are unavoidable meaning vulnerabilities on the vendor side from the complex nature of multi-tenant environments, or even security vulnerabilities from other tenants can cause critical security issues for the organization. A lack of logical isolation from the vendor between its tenants can lead to security breaches the tenant is unable to control spreading catastrophically. From the organization's side since they are responsible for securing the platform and application on their end misconfigurations can become more likely from complex customized systems leading to further security risks. An issue shared with PaaS, IaaS reliant organizations can also suffer from vendor lock-in, as moving between service vendors can be difficult and time consuming as it often involves bringing daily operations and business activities offline for extended periods. Lastly, a disadvantage shared by all these primary cloud models, regulatory and legal adherence can be issues especially when servicing geographically diverse and multinational organizations that may adhere to conflicting legal requirements or have a complex array of regulations that need to be maintained.

Figure 2



“Cloud Computing Models SaaS, IaaS, and PaaS” V-Soft Consulting

Cloud Architecture and Services

AWS and its many tools and services offered can fall under categorization as SaaS, PaaS, and IaaS. Thus, for this section examining tools and architecture AWS will be examined for specific examples and references towards answering the initial research questions. AWS is supported by 66 Availability Zones (AZ) with 21 geographically distributed regions. This virtualized IT infrastructure is secure, cost effective and puts developers in full control of instances.

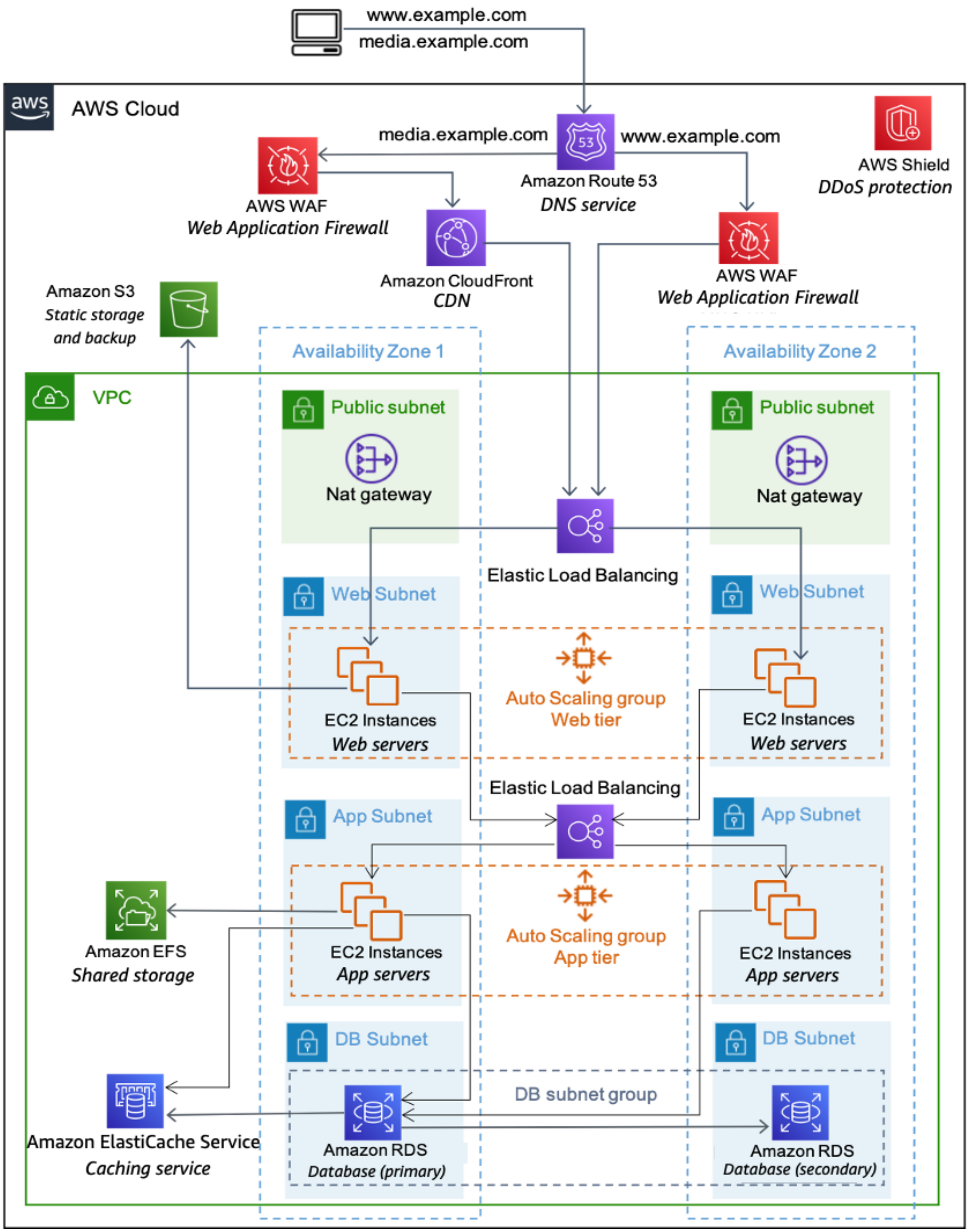
Amazon Elastic Cloud Computing (EC2) provides easy to use tools and applications to facilitate this infrastructure and computing capabilities. Virtual private clouds and other network segmentation features allow for in depth customization for powerful and secure network configuration. Amazon Elasticsearch handles load balancing and instance consistency while reducing operational overhead. AWS Relational Database Services (RDS) provides the tools for virtualized database configuration and deployment while also offering resiliency with replication in multiple availability zones. RDS also provides automation as failed instances are automatically detected and replaced automatically. These tools and services support the deployment and management of virtualized infrastructure and database platforms that tenant companies rely on while also ensuring they are highly available and flexible to meet changing needs.

In addition to these benefits, from a security perspective AWS offers several services that can enhance layered security in the cloud and help with authentication and authorization. Amazon Cognito is one of these services that offers integral features like multifactor authentication (MFA) and identity federation with other external identity providers.

AWS Identity and Access Management (IAM) allows management of internal security features. This includes defining and controlling roles, groups, and security permissions. IAM also provides MFA and a centrally managed platform for AWS accounts and applications. Along with these access control services features like AWS Key Management System (KMS) allow for the creation and utilization of encryption keys to secure data. KMS is integrated into EC2 and RDS and other cloud processes while monitored by auditing services like AWS CloudTrail and CloudWatch.

For SaaS cases AWS Backup exists for reliability as it can automate the backup and saving process ensuring that all progression on a software application is saved for a user. IAM and CloudWatch give users better access control methods and secluded secure service delivery. AWS also supports PaaS tenants with services like AWS Elastic Beanstalk. Elastic Beanstalk offers a stage for developers to code test and deploy applications, it supports a variety of languages and frameworks. Additionally, the backend services such as resource provisioning, scaling, load balancing, and security monitoring are all handled by Beanstalk and do not require overhead from the developer company. Another PaaS styled service that AWS offers is AWS Lambda. Lambda is a robust service that enables developers to utilize various tools of the AWS ecosystem for software development and deployment. Notably, this service is serverless allowing the execution of code without on-premises infrastructure. Lambda functions are what manage the actual execution of code and operate in an efficient manner as resources for these processes are allocated automatically and only utilized for runtime, thus making it very economical. Lambda APIs allow a wide degree of customization for streamlined integration of other tools into program design on the platform.

Figure 3



https://docs.aws.amazon.com/whitepapers

Research Solutions and Conclusion

After detailing the various main cloud models – SaaS, PaaS, and IaaS, along with their advantages, disadvantages, and potential security risks, it becomes evident that cloud computing offers a range of solutions for organizations of all sizes. AWS offers a comprehensive array of services to implement and support the deployment of those cloud models. Circling back to the initially proposed research questions, the insight gained from this can be used to propose effective solutions and mitigation strategies.

Firstly, the initial research question was about how to maintain confidentiality and integrity of data that is stored in a cloud infrastructure. AWS offers a variety of services that companies can leverage to solve this issue. AWS KMS allows for encryption key issuing and management that helps keep data stored on the cloud and data in transit confidential and secure. Additionally, AWS IAM is one of the access control methods that allows organizations to create, combine, and manage roles, groups and security permissions in any infrastructure configuration AWS offers. CloudWatch is an integrated monitoring service that monitors applications and infrastructure and can even set alerts for unauthorized changes and activity ensuring integrity.

The second research question regarding how to maintain data compliance and regulatory adherence is also solved with AWS services and management. Ultimately the geographically distributed nature of cloud infrastructure and the powerful streamlined tools offered to deploy various services let organizations ensure data compliance to their local regulations in configurations where it is not already handled by AWS. AWS CloudTrail and CloudWatch are two of these services that ensure compliance and offer data auditing.

The final research question proposed in the introduction of this paper was how to harden cloud infrastructure, mitigate cyberattacks and the availability concerns that result from them. This was an important question to pose as cloud vendors by operation support the critical business-related goals and infrastructure for all their tenants. Thus, maintaining high availability of these resources that tenant organization depend on is of critical importance. The solution to this comes down to a combination of intelligent and robust designed infrastructure with powerful security monitoring tools. The globally distributed infrastructure of AWS with its regions and AZ subdivisions combined with redundancy organization of data storage led to a very robust cloud environment with the highest availability among cloud vendors (SOURCE). This combined with powerful monitoring tools like CloudWatch and CloudTrail and Elastic Load Balancing for traffic demands helps keep the infrastructure highly scalable and monitored for security incidents.

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